

In[196]:=

```
printShellSatSI[satarea_, albedo_, range_, band_, angle_,
  aperD_, photD_, itime_] := Module[{signal, noise, sky, photRate},
  signal =
    itime ×
    photonsSI[
      shellSatRadIntSI[satarea, albedo, range, band, angle] ×
       $\frac{\pi (aperD / 2)^2}{range^2}$ ,
      wavelengthSI[band] ];
  sky = itime ×
    photonsSI[
      skyfobservatorySI[
        wavelengthSI[band]] ×
        widthSI[band] ×
         $\pi (aperD / 2)^2 \pi (photD / 2)^2$ ,
        wavelengthSI[band] ];
  Print[signal, " = signal ", sky, " = sky ",
    Sqrt[sky], " = sqrt sky"];
  signal / Sqrt[signal + sky]
];
```

In[197]:=

```
printShellSatSI[
   $\pi$ ,
  0.2,
  100 000 000,
  "V",
  0,
  0.5,
  1 arcsec,
  10
]
8337.98 = signal 31.3976 = sky 5.60336 = sqrt sky
```

Out[197]=

```
91.1411

8337.98 31.3976 91.1411
8337.98 31.3976 91.1411
```

Out[192]=

```
91.1411
```

In[185]:=

arcsec

Out[185]=

$$\frac{\pi}{648\,000}$$